HANWEN GUO

GUO@HANWEN.IO HANWEN.IO GITHUB GOOGLE SCHOLAR

EDUCATION

University of Utah, Ph.D. in Computer Science 08/2024 – 12/2029 (expected)
Advisor: Ben Greenman | Research Focus: Programming Languages, Type Systems, Gradual Typing, Program Analysis

Wuhan University, B.Eng. in Computer Science Fall 2020 – Spring 2024

RESEARCH EXPERIENCE

PPG: Automated Auditing for Python TypeGuard Predicates Spring 2025 – Present

- Designed PPG, a hybrid static-analysis and property-based testing approach for auditing the semantic validity of Python `TypeGuard` predicates by extracting `True`-returning path constraints and synthesizing acceptance-directed Hypothesis tests.
- Audited 308 TypeGuard predicates across 60 open-source projects (>5M lines of Python), finding 50 reproducible counterexamples across 23 projects and characterizing recurring sources of unsoundness and barriers to automated validation.

If-T: A Benchmark for Type Narrowing Fall 2024 – Spring 2025

- Proposed and implemented a language-agnostic benchmark with 13 scenarios spanning three fundamental dimensions of type-narrowing design.
- Evaluated 10 gradual type checkers (including TypeScript, Flow, mypy, Pyright, ty, Pyrefly, Sorbet, Luau) and analyzed differences in narrowing precision and supported type-narrowing designs.
- The benchmark was later used and credited by the Elixir team in evaluating Elixir's type system.

STIR: Statistical Type Inference for Incomplete Programs Fall 2023 – Spring 2024

- Contributed to the research artifact by restructuring benchmark workflows into a modular, parameterized architecture and optimizing the evaluation pipeline through profiling, parallelization, and C++ reimplementation of data-processing components.

PUBLICATIONS

[1] Hanwen Guo, Ben Greenman. 2025. If-T: A Benchmark for Type Narrowing. *<Programming>* 10.2. *Editors' Choice Award.* [CODE] [DOI] [PDF] [POSTER]

[2] Y. Peng et al. 2023. Statistical Type Inference for Incomplete Programs. *ESEC/FSE 2023. Best Artifact Award.* [CODE] [DOI] [PDF]

TECHNICAL SKILLS

- Languages: Python, C++

TEACHING & SERVICE

Artifact Evaluation Committee, <Programming> 11 Spring 2026

Teaching Assistant, University of Utah Fall 2025, Spring 2026, Fall 2026
Programming Language, Operating Systems